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CS101

From pseudocode to Python

Today
Welcome to your next lesson: From pseudocode to Python.

I'm getting your personalized lesson ready—based on what you've learned, what's coming next, and the best way to help you grow.

Step by step, you’re moving forward.

Ron, let’s make this really concrete.

Imagine this pseudocode for a shopping cart 🛒:

```
CHECK if cart has any items

IF cart is empty

  SHOW "Your cart is empty"

ELSE

  SHOW "You have X items in your cart"
```

Right now this is just pseudocode — a plan in plain English. Python can’t run it yet. Our job in this lesson is to turn each line into real Python, one line at a time, while keeping the same structure.

To start simple: in your own words, how would you describe the difference between pseudocode and Python code?

You can keep it short, like: “Pseudocode is …, Python is …”

pseudocode describes the flow of what we want to code ,in plain english
python code is th actual executable instructions to run it

Exactly right 🎯

Pseudocode: describes the flow in plain English (the recipe).
Python: the exact executable instructions the computer can run (the cooking).
Now we’ll practice turning that recipe into real code, line by line.

Here’s the same shopping-cart pseudocode again:

```
CHECK if cart has any items

IF cart is empty

```

□□SHOW "Your cart is empty"

ELSE

□□SHOW "You have X items in your cart"

Step 1 (I'll guide):

Let's assume cart_items is a Python list holding the items.

The English idea "CHECK if cart has any items" could become something like:

Python

if len(cart_items) == 0:
I'll explain why later — for now, just focus on:

pseudocode condition → Python if with a condition.

Does mapping that first line to an if with a condition make sense so far, or does any part feel fuzzy?

That makes sense

Great 👍

Let's go one small step further and keep it very visual.

Here's a side□by□side "ladder" for the first two pseudocode lines:

Pseudocode	Python idea (not full code yet)
CHECK if cart has any items	if len(cart_items) == 0:
IF cart is empty	(this is exactly the same condition)

So in this case, the first two lines of pseudocode both point to one Python if line: we're "checking" and we're saying what we check for (empty cart).

Next step, we need to translate this pseudocode line:

SHOW "Your cart is empty"

In Python, that "SHOW" is usually a print(...).

So this might become:

Python

print("Your cart is empty")
But this line must be indented under the if so Python knows it belongs to that if.

Quick check: are you comfortable with the idea that

the if line ends with :
and the print line is indented under it
to show they're connected?

If yes, I'll have you translate the ELSE part next.

Yes, I'm comfortable

What do you think? Pick the option that matches the pseudocode’s indentation best 👇

Look at this pseudocode and Python. Which Python version correctly matches the indentation structure?

Pseudocode:

IF cart is empty □□SHOW "Your cart is empty" ELSE □□SHOW "You have items"

Option A:

```
if len(cart_items) == 0:
print("Your cart is empty")
else:
print("You have items")
```

Option B:

```
if len(cart_items) == 0:
    print("Your cart is empty")
else:
    print("You have items")
```

A.

Option A

B.

Option B

Nice job, Ron 🎉 You picked Option B, which is exactly right.

Why it’s correct, step by step:

The if and else lines are not indented (they’re the main structure).

Both print lines are indented the same amount, showing they belong inside their if or else block.

That matches the pseudocode structure:

Under IF cart is empty → one indented action
Under ELSE → one indented action
Now let’s flip control to you 💻

Based on the full pseudocode:

CHECK if cart has any items

IF cart is empty

□□SHOW "Your cart is empty"

ELSE

□□SHOW "You have X items in your cart"

Here’s what I’d like you to do:

Open the Python code editor for this lesson.

Create a variable cart_items with a list, for example: cart_items = [] (empty for now).

Write Python code that:

Checks if the cart is empty.
Prints "Your cart is empty" if it is.
Prints "You have X items in your cart" if it isn't, where X is the number of items (hint: len(cart_items)).
Hit Run to test it, then Submit code so I can see it.

Don't worry about being perfect; I'll help you fine-tune it after I see your code.

Python

```
cart_items = []

if len(cart_items) == 0:
    print("Your cart is empty")
else:
    print(f"You have {len(cart_items)} items in your cart")
```

Ron, that's a clean translation 👏

You did exactly what this lesson is about:

Kept the structure: if / else with proper indentation.
Matched the meaning:
Empty cart → "Your cart is empty"
Non-empty cart → message with len(cart_items) plugged in.
Used an f-string nicely: f"You have {len(cart_items)} items in your cart".

This is a perfect example of pseudocode → Python, line by line.

One more tiny check to lock this in 🔒:

Try this in your editor (don't submit yet, just run it):

Python

```
cart_items = ["apple", "banana"]

if len(cart_items) == 0:
    print("Your cart is empty")
else:
    print(f"You have {len(cart_items)} items in your cart")
```

Then tell me:

What output do you see when cart_items has those two items?

had done that already testing it

Nice — love that you tested it already 🔁😂

What you just did is the full translation loop:

Start from pseudocode (the plan).
Map each step to Python, keeping structure and indentation.
Run it with different inputs to confirm the behavior matches the plan.

That's the mastery outcome for this lesson. 🎯

Before we officially close it out:

Would you like to do one more quick pseudocode → Python mini-example together, or are you feeling ready to wrap up this lesson and move on to the next one?

Let's do one more example

Love that. One more example is a great way to lock this in 👍

Here’s a new pseudocode recipe:

```
ASK user for a number

IF number is greater than 10

□□SHOW "Big number"

ELSE

□□SHOW "Small or equal to 10"
```

We’ll keep the same idea: one pseudocode step → one Python step, with structure preserved.

Your turn in the code editor:

Use input() to ask the user for a number.
Convert it to an int (so you can compare it).
Write an if / else:
If number > 10 → print "Big number".
Else → print "Small or equal to 10".

Run it with a few values (like 5 and 15), then hit Submit code so I can review it.

Python

```
number = int(input("Please enter a number: "))
if number > 10:
    print("Big number")
else:
    print("Small or equal to 10")
```

Beautifully done, Ron 🎯

You just:

Took clear pseudocode (“ASK… IF… ELSE…”)
Translated it line by line into Python
Preserved the structure with correct indentation
Used input + comparison exactly as planned
This is exactly the skill for this lesson: turning a plan into working Python without losing the logic.

Quick recap of your translation pattern:

ASK user for a number → input(...) + int(...)
IF number is greater than 10 → if number > 10:
SHOW "Big number" → indented print("Big number")
ELSE / SHOW "Small or equal to 10" → else: + indented print(...)
That’s your reusable pseudocode → Python muscle. 👍

You’re ready for the next lesson.

When you’re set, click Next lesson in the Learn tab, and we’ll keep building on this.

```
cart_items = ["apple", "banana"]

if len(cart_items) == 0:
    print("Your cart is empty")
else:
    print(f"You have {len(cart_items)} items in your cart")

----------------------

number = int(input("Please enter a number: "))
if number > 10:
    print("Big number")
else:
    print("Small or equal to 10")
```